R16

Code No: 135BY

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, November/December - 2018
COMPUTER GRAPHICS
(Common to CE, CSE, IT)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A

(25 Marks)

1.a)	List some of the input devices.	[2]
b)	Write the drawbacks of scan line algorithm.	[3]
c)	Write the properties of homogeneous coordinates.	[2]
d)	Explain about 3D shear transformation.	[3]
e)	What is quadratic surface?	[2]
f)	Write the applications of Bezier splines.	[3]
g)	What is transparency?	[2]
h)	What are the advantages of back face detection method?	[3]
i)	What are the applications of computer animation?	[2]
j)	Explain about direct motion specification.	[3]

PART - B

(50 Marks)

2.a)	Explain DDA scan algorithm in detail.	
b)	Compare raster scan system and random scan system.	[5+5]
	OR	
3.a)	Write midpoint ellipse drawing algorithm.	
b)	Explain in detail the working principle of cathode ray tube.	[5+5]
4.a)	Write Sutherland Hodgeman algorithm.	
b)	Prove that two rotation matrices are commute $R_1R_2 = R_2R_1$.	[5+5]
	OR	
5.a)	Write Cyrus-beck line clipping algorithm.	
b)	Explain about composite transformation.	[5+5]
6.a)	Explain 3D composite transformations for scaling, rotation.	
b)	Derive the transformation matrix for Hermite curve.	[5+5]
	OR	
7.a)	Compare parallel and perspective projections.	
b)	Explain Bezier curves in detail.	[5+5]

user paper 16-12-2018PM

8.a) Explain z-buffering method for hidden surface detection.
b) Explain about the specular reflection and the phong model. [5+5]

OR

9.a) Explain about gourmand shading algorithm.
b) Give a note on octree method for visible surface detection. [5+5]

10.a) Describe method to create image using iterated functions.
b) Explain raster animation in detail. [5+5]

OR

11.a) Explain key frame system in detail.
b) Explain about the design of animation sequences. [5+5]

---ooOoo---

Used paper 16-12-2018PM